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Large-aperture compact scanning tele cameras

Abstract

Scanning Tele cameras (STCs) based on two optical path folding element (OPFE) field-of-view scanning and mobile devices including such STCs. A STC may comprise a first OPFE (O-OPFE) for folding a first optical path OP1 to a second optical path OP2, an O-OPFE actuator, a second OPFE (I-OPFE) for folding OP2 to a third optical path OP3, an I-OPFE actuator, a lens, a lens actuator and an image sensor, wherein the STC has a STC native field-of-view (n-FOV.sub.T), wherein the O-OPFE actuator is configured to rotate the O-OPFE around a first axis and the I-OPFE actuator rotates the I-OPFE around a second axis for scanning a scene with the n-FOV.sub.T, wherein the lens actuator is configured to move the lens for focusing along a third axis, and wherein the first axis is perpendicular to the second axis and parallel to the third axis.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation from U.S. patent application Ser. No. 19/000,742 filed Dec. 24, 2024 (now allowed), which was a continuation from U.S. patent application Ser. No. 18/718,461 filed Jun. 11, 2024 (issued as U.S. Pat. No. 12,216,259), which was a 371 application from international patent application PCT/IB2022/060748 filed Nov. 8, 2022, which claims the benefit of priority from U.S. Provisional patent applications Nos. 63/289,323 filed Dec. 14, 2021, 63/297,256 filed Jan. 7, 2022, and 63/380,786 filed Oct. 25, 2022, all of which are incorporated herein by reference in their entirety.

FIELD

(1) The subject matter disclosed herein relates in general to compact mobile cameras and in particular to mobile scanning telephoto (“Tele”) cameras.

DEFINITIONS

(2) In this application and for optical and other properties mentioned throughout the description and figures, the following symbols and abbreviations are used, all for terms known in the art: Lens element: a single lens element. Lens: assembly of a plurality of lens elements. Total track length (TTL): the maximal distance, measured along an axis parallel to the optical axis of a lens, between a point of the front surface S.sub.1 of a first lens element L.sub.1 and an image sensor, when the system is focused to an infinity object distance. Back focal length (BFL): the minimal distance, measured along an axis parallel to the optical axis of a lens, between a point of the rear surface

S.sub.2N of the last lens element L.sub.N and an image sensor, when the system is focused to an infinity object distance. Effective focal length (EFL): in a lens (assembly of lens elements L.sub.1 to L.sub.N), the distance between a rear principal point P' and a rear focal point F' of the lens. f-number (f/#): the ratio of the EFL to an entrance pupil (or aperture) diameter of a lens.

BACKGROUND

- (3) Mobile electronic handheld devices (or just “mobile devices”) such as smartphones having two or more compact cameras (also referred to as “multi-cameras”) are known. The two or more cameras have lenses with different effective focal lengths (EFLs) that capture images of a same scene with different fields of view (FOVs). For example, a multi-camera may include a Wide camera having a Wide camera FOV (“FOV.sub.W”) of e.g. 80 degrees and a Tele (or “zoom”) camera having a narrower FOV (“native FOV.sub.T” or “n-FOV.sub.T”) of e.g. 25 degrees and with higher spatial resolution (for example 3-5 times higher) than that of the Wide camera.
- (4) Tele cameras with scanning capability (“scanning Tele cameras” or “STCs”) for expanding the native fields-of-view n-FOV.sub.T to an effective Tele FOV (also referred to as “scanning FOV.sub.T” or “s-FOV.sub.T”) overcome some of the limitations that relate to narrow n-FOV.sub.Ts. Compact STCs can be realized in a folded camera such as described for example in co-owned U.S. Pat. No. 10,578,948, or in a double-folded camera such as described for example in co-owned international patent PCT/IB2021/059843. One or two optical path folding elements (OPFEs), for example prisms or mirrors, are rotated along one or two directions to direct (or “scan” or “steer”) the n-FOV.sub.T towards arbitrary points of view (POVs) within s-FOV.sub.T.
- (5) The f-number (“f/#”) of a camera lens is the ratio of the EFL to the aperture diameter (“DA”): $f/\# = \text{EFL}/\text{DA}$. As known in the art, a low f/# is desired as of 3 major advantages: High signal-to-noise ratio (SNR), strong “natural” Bokeh effect and support of high image resolutions. A low f/# is, amongst others, achieved by maximizing the aperture area of the camera lens.
- (6) It is noted that herein, “aperture” refers to an entrance pupil of a lens (or “lens assembly”). If it is referred to an “aperture of a camera” or an “aperture of an optical lens system”, this always refers to the aperture of the lens included in the camera or in the optical lens system respectively. “Aperture” and “clear aperture” are used interchangeably.
- (7) There is need and it would be beneficial to have a compact scanning Tele camera for incorporation in a mobile device that includes a lens having a large aperture area and low f number.

SUMMARY

- (8) In various exemplary embodiments (examples), there is provided a scanning Tele cameras (STC) comprising: an OPFE for folding a first optical path OP1 to a second optical path OP2, wherein the OPFE has an OPFE height H.sub.O measured along OP1, an OPFE length L.sub.O measured along an axis parallel to OP, and a light exiting surface; an OPFE actuator; a lens having a lens optical axis parallel to OP2, an EFL, a maximum lens aperture height H.sub.A measured along OP1, and a maximum lens aperture width W.sub.A measured along an axis perpendicular to both OP1 and OP2; and an image sensor having an image sensor diagonal (SD) and an image sensor height H.sub.sensor measured along OP1, wherein the STC has a STC native field-of-view (n-FOV.sub.T), wherein the OPFE actuator is configured to rotate the OPFE around a first rotation axis perpendicular to both OP1 and OP2 and around a second rotation axis parallel to OP1 for scanning a scene with the n-FOV.sub.T, wherein the first rotation axis is located at a distance $\Delta 1$ from the light exiting surface of the OPFE, and wherein $\Delta 1/L_{\text{sub.O}} < 0.25$.
- (9) In some examples, $\Delta 1/L_{\text{sub.O}} < 0.2$. In some examples, $\Delta 1/L_{\text{sub.O}} < 0.15$. In some examples, $\Delta 1/L_{\text{sub.O}} < 0.1$. In some examples, $\Delta 1/L_{\text{sub.O}} < 0.075$.
- (10) In some examples, the OPFE has an OPFE center with respect to OP1, wherein the first rotation axis is located at a distance $\Delta_{\text{sub.C}}$ from the OPFE center along OP1, and wherein a ratio of $\Delta_{\text{sub.C}}$ and H.sub.O fulfils $\Delta_{\text{sub.C}}/H_{\text{sub.O}} > 0.015$. In some examples, $\Delta_{\text{sub.C}}/H_{\text{sub.O}} > 0.02$.
- (11) In some examples, the lens is movable for focusing along OP2.
- (12) In some examples, the STC is included in a camera module, wherein the camera module is

divided into a module region having a module region height $H_{sub.M}$ and a shoulder region having a shoulder region height $H_{sub.S} < H_{sub.M}$, all heights being measured along OP1, and wherein $H_{sub.S} < H_{sub.A} + 3$ mm. In some examples, the OPFE is included in the module region and the lens and the image sensor are included in the shoulder region.

(13) In some examples, the lens is divided into a first lens group (G1) and a second lens group (G2), wherein the OPFE and G1 are included in the module region and G2 and the image sensor are included in the shoulder region.

(14) In some examples, the OPFE has an OPFE center with respect to OP1, wherein the first rotation axis is located at a distance $\Delta_{sub.C}$ from the OPFE center along OP1, and a ratio of $\Delta_{sub.C}$ and $H_{sub.S}$ fulfils $\Delta_{sub.C}/H_{sub.S} > 0.01$. In some examples, $\Delta_{sub.C}/H_{sub.S} > 0.015$.

(15) In some examples, $H_{sub.S} < H_{sub.A} + 2$ mm.

(16) In some examples, $H_{sub.S} < W_{sub.A}$.

(17) In some examples, $H_{sub.S}/H_{sub.M} < 0.9$. In some examples, $H_{sub.S}/H_{sub.M} \leq 0.8$.

(18) In some examples, $H_{sub.A}/H_{sub.S} > 0.7$.

(19) In some examples, $H_{sub.A}/H_{sub.M} > 0.5$.

(20) In some examples, $DA/H_{sub.S} > 0.8$. In some examples, $DA/H_{sub.M} > 0.65$.

(21) In some examples, $H_{sub.M} < H_{sub.O} + 4$ mm. In some examples, $H_{sub.M} < H_{sub.O} + 3$ mm.

(22) In some examples, the STC is included in a mobile device, wherein the mobile device has a regular region with a regular thickness T and a bump region with a bump thickness T+B, wherein the shoulder region is included in the mobile device regular region and wherein the module region is included in the mobile device bump region. In some examples, the mobile device may additionally include a Wide camera having a Wide camera image sensor and a Wide camera field-of-view (FOV_{sub.W}). In some examples, the mobile device may be a smartphone.

(23) In some examples, $H_{sub.O} < H_{sub.A} + 2$ mm. In some examples, $H_{sub.O} < H_{sub.A} + 1$ mm.

(24) In some examples, $SD/EFL > 0.4$ mm.

(25) In some examples, the STC uses a parallel STC sensor configuration. In some examples, the STC uses an anti-parallel STC sensor configuration.

(26) In some examples, the scanning provides an effective Tele scanning FOV s-FOV_{sub.T}, wherein the s-FOV_{sub.T} has a longer horizontal side and a shorter vertical side, and wherein a horizontal side H-FOV_{sub.T} of s-FOV_{sub.T} is greater than 40 degrees.

(27) In some examples, a center location of s-FOV_{sub.T} is identical with a center location of FOV_{sub.W}. In some examples, FOV_{sub.W} is in the range of 50-120 degrees, wherein s-FOV_{sub.T} covers a 16:9 segment of FOV_{sub.W}. In some examples, FOV_{sub.W} is in the range of 70-90 degrees, wherein s-FOV_{sub.T} covers a 16:9 segment of FOV_{sub.W}. In some examples, FOV_{sub.W} is in the range of 75-85 degrees, wherein s-FOV_{sub.T} covers a 16:9 segment of FOV_{sub.W}.

(28) In some examples, H-FOV_{sub.T} > 45 degrees. In some examples, H-FOV_{sub.T} > 50 degrees.

(29) In some examples, s-FOV_{sub.T} has a longer horizontal side and a shorter vertical side, and a vertical side V-FOV_{sub.T} of s-FOV_{sub.T} is greater than 20 degrees. In some examples, V-FOV_{sub.T} > 25 degrees. In some examples, V-FOV_{sub.T} > 30 degrees.

(30) In some examples, the rotation of the OPFE along the first rotation axis is by more than ± 5 degrees around a zero scan position. In some examples, the rotation of the OPFE along the second rotation axis is by more than ± 15 degrees around a zero scan position.

(31) In some examples, the OPFE is a prism. In some examples in which the OPFE is a prism, the prism has a fast scanning axis and a slow scanning axis, wherein the image sensor is oriented such that the prism's fast scanning axis is aligned with the horizontal side H-FOV_{sub.T} of S-FOV_{sub.T}.

(32) In some examples, the STC has an EFL of 8-10 mm. In some examples, the STC has an EFL of 10-25 mm. In some examples, the STC has an EFL of 25-50 mm.

(33) In some examples, the OPFE actuator is a voice coil motor. In some examples, the lens has a f number f/#, wherein $f/\# < 3.5$. In some examples, $f/\# < 3$. In some examples, $f/\# < 2.5$. In some

examples, a distance between the OPFE and the lens is $\Delta L_{sub.O}$, and $\Delta L_{sub.O}/TTL < 0.25$.

(34) In some examples, $H_{sub.O}/L_{sub.O} < 0.9$. In some examples, $W_{sub.O}/H_{sub.O} > 1.5$. In some examples, $W_{sub.O}/H_{sub.O} > 1.75$.

(35) In some examples, the lens is a cut lens and the cut is performed along an axis parallel to OP2. In some examples, the cut lens is cut by 10%-50%. In some examples, the cut lens is cut by X %, wherein the cutting by X % reduces $MH_{sub.M}$ and $MH_{sub.S}$ by $0.5 \times X \% - X \%$.

(36) In some examples, the OPFE is a cut OPFE and the cut is performed along an axis parallel to OP2. In some examples, the cut OPFE is cut by 10%-40%. In some examples, the cut OPFE has a cut surface that extends along a distance Δ_{cut} from the OPFE light exiting surface, wherein $\Delta_{cut} > 41$.

(37) In some examples, the lens elements in the lens have an average lens thickness (ALT), the thickness of the first lens element $L_{sub.1}$ being $T1$, and $T1/ALT > 1.5$.

(38) In some examples, the lens elements in G1 have an ALT of $ALT_{sub.G1}$, and $ALT_{sub.G1}/ALT > 1.25$.

(39) In some examples, the lens elements in G2 have an ALT of $ALT_{sub.G2}$, and $ALT_{sub.G1}/ALT_{sub.G2} > 2$.

(40) In some examples, the focal length of the first lens element is $f1$ and $f1/EFL < 0.75$. In some examples, a ratio of the height of G1 and the height of G2 is $H_{sub.G1}/H_{sub.G2} > 1.15$.

(41) In some examples, $H_{sub.G1}/H_{sub.G2} > 1.3$.

(42) In some examples, the first lens element L1 is made of glass.

(43) In some examples in which the OPFE is a prism, the prism includes a stray light prevention mechanism. In some examples, the stray light prevention mechanism includes two stray light masks located at the light entering surface and two stray light masks located at the light exiting surface. In some examples, the two stray light masks located at the light entering surface are located at a left margin and at a right margin of the light entering surface, and wherein the two stray light masks located at the light exiting surface are located at a top and at a bottom of the light exiting surface. In some examples, the two stray light masks located at the light entering surface together cover a surface area of more than 10% and less than 20% of the area of the light entering surface. In some examples, the two stray light masks located at the light exiting surface together cover a surface area of more than 20% and less than 30% of the area of the light exiting surface.

(44) In some examples there is provided a mobile device comprising an STC as above or below, together with an application processor (AP). In some examples, in a mobile device that includes a Wide camera, the AP is configured to use image data from the Wide camera for autonomous scanning of a scene with the STC's $n\text{-FOV}_{sub.T}$. In some examples, the AP is configured to scan a scene with the STC's $n\text{-FOV}_{sub.T}$ according to a user input.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein, and should not be considered limiting in any way. Like elements in different drawings may be indicated by like numerals. Elements in the drawings are not necessarily drawn to scale.
- (2) FIGS. 1A-1D show known dual cameras including a folded STC included in a mobile device;
- (3) FIG. 1E shows an OPFE of a known STC as in FIGS. 1A-1D in a zero scan position in a cross-sectional view;
- (4) FIG. 2A shows a folded STC disclosed herein in a cross-sectional view;
- (5) FIG. 2B shows a folded STC disclosed herein in a top view;

- (6) FIG. 2C shows another folded STC disclosed herein in a cross-sectional view;
- (7) FIG. 2D shows another folded STC disclosed herein in a top view;
- (8) FIG. 3A shows a mobile device including the folded STC from FIGS. 2A-B in a cross-sectional view;
- (9) FIG. 3B shows another mobile device including the folded STC from FIGS. 2C-D in a cross-sectional view;
- (10) FIG. 3C shows yet another mobile device including a folded camera of FIGS. 2A-D and a Wide camera in a perspective view;
- (11) FIG. 3D shows the mobile device of FIG. 3C in a top view;
- (12) FIG. 3E shows the mobile device of FIG. 3C in a side view;
- (13) FIG. 3F shows yet another mobile device including a STC disclosed herein as well as a Wide camera;
- (14) FIG. 3G shows the mobile device if FIG. 3F in a top view;
- (15) FIG. 3H shows the mobile device if FIG. 3F in a side view;
- (16) FIGS. 4A-4C show the folded STC from FIGS. 2A-B at different scan states around a first rotation axis in a cross-sectional view;
- (17) FIGS. 4D-4F show the folded STC from FIGS. 2C-D at different scan states around a first rotation axis in a cross-sectional view;
- (18) FIGS. 5A-5C show the folded STC from FIGS. 2A-B at different scan states around a second rotation axis in a cross-sectional view;
- (19) FIGS. 5D-5F show the folded STC from FIGS. 2C-D at different scan states around a second rotation axis in a cross-sectional view;
- (20) FIGS. 6A and 6B shows OPFE of the folded STC from FIGS. 2A-B and the first and the second rotation axis;
- (21) FIGS. 6C and 6D shows OPFE of the folded STC from FIGS. 2C-D and the first and the second rotation axis;
- (22) FIG. 7 shows a field-of-view of the folded STC from FIGS. 2A-B and FIGS. 2C-D;
- (23) FIG. 8 shows an optical lens system included in the folded STC from FIGS. 2A-B.
- (24) FIG. 9 shows an optical lens system included in the folded STC from FIGS. 2C-D.
- (25) FIG. 10 shows another optical lens system disclosed herein;
- (26) FIG. 11A shows yet another optical lens system disclosed herein in a cross-sectional view;
- (27) FIG. 11B shows yet another optical lens system disclosed herein in a top view;
- (28) FIG. 11C shows yet another optical lens system disclosed herein in a perspective view;
- (29) FIG. 12 shows a mobile device including a STC which includes the yet another optical lens system of FIGS. 11A-C;
- (30) FIG. 13A shows a prism included in the yet another optical lens system of FIGS. 11A-C in a side view;
- (31) FIG. 13B shows the prism of FIG. 13A in a perspective view;

DETAILED DESCRIPTION

(32) FIG. 1A illustrates a known folded scanning Tele camera (“folded STC”) **100** comprising an optical path folding element (OPFE) **102**, a lens **104** including a plurality of lens elements (not shown), lens **104** being included in a lens barrel **110**, and an image sensor **106**. Lens **104** has an optical lens height $H_{\text{sub.L}}$, measured along OP **112**. $H_{\text{sub.L}}$ defines an aperture diameter (DA) of lens **104** in the y-direction. OPFE **102** folds an optical path (OP) from a first OP **112** (parallel with the y-axis in the YZ coordinate system shown) to a second OP **108** parallel with an optical axis of lens **104** along the z axis in the coordinate system shown. A theoretical limit for a height of a camera module (“minimum module height” or “ $MH_{\text{sub.M}}$ ”) and a theoretical limit for a length of a camera module (“minimum module length” or “ $ML_{\text{sub.M}}$ ”) including camera **100** is shown. $MH_{\text{sub.M}}$ and $ML_{\text{sub.M}}$ are defined by the largest dimension along OP **112** and along OP **108** of a component included in camera **100** respectively. For scanning a scene with STC **100**'s n-

FOV.sub.T, OPFE **102** may be rotated around two axes, a first rotation axis being parallel to the y-axis, and a second rotation axis being parallel to the x-axis.

(33) FIG. **1B** illustrates a known dual-camera **150** that comprises STC **100** and a (vertical or upright) Wide camera **130** including a Wide lens **132** and a Wide image sensor **138**. Lens **132** is included in a lens barrel **134**. Wide camera **130** has an OP **136** which is substantially parallel with OP **112**.

(34) FIG. **1C** shows schematically a known mobile device **160** (e.g. a smartphone) having an exterior rear surface **162** and including known STC **100** in a cross-sectional view. The aperture of STC **100** is located at rear surface **162**, a front surface **166** may e.g. include a screen (not visible). Mobile device **160** has a regular region of thickness (“T”) and a camera bump region **164** that is elevated by a height B over the regular region. The bump region has a bump length (“L.sub.B”) and a bump thickness T+B. As shown, STC **100** is entirely integrated in the bump region, such that MML and MMH define a lower limit for the bump region, i.e. for L.sub.B and T+B. For industrial design reasons, a small camera bump (i.e. a short L.sub.B) is desired. Mobile device **160** may additionally include an application processor (AP—not shown). In some examples, the AP may be configured to scan a scene with STC **100**'s n-FOV.sub.T according to a user input. In other examples, the AP may be configured to use image data from a Wide camera such as camera **130** to autonomously scan a scene with STC **100**'s n-FOV.sub.T.

(35) FIG. **1D** shows a known cut lens element **180** in a cross-sectional view. Lens element **180** is cut by 20%, i.e. **180**'s optical width W.sub.L is 20% larger than its optical height H.sub.L. This means that also the aperture changes accordingly, such that the aperture is not axial symmetric. The cutting allows for a small H.sub.L, which is required for small MH.sub.Ms (see FIG. **1A**), and still relatively large effective aperture diameters (DAs) which satisfy $DA > H.sub.L$.

(36) FIG. **1E** shows OPFE **102** of known STC **100** in a zero scan position in a cross-sectional view. Here, OPFE **102** is a prism having a light entering surface **190** and a light exiting surface **191**. In the zero scan position, OPFE **102** has a length L.sub.O (measured along the z-axis) and a height H.sub.O (measured along the y-axis). The location of a first rotation axis **192** for rotating OPFE **102** around an axis perpendicular to the shown y-z-coordinate system (and as indicated by arrow **193**) and a second rotation axis **194** for rotating OPFE **102** around an axis parallel to the axis (and as indicated by arrow **195**) are shown. A distance from first rotation axis **192** to OPFE **102**'s light exiting surface **191** is marked $\Delta 192$. A distance from second rotation axis **194** to OPFE **102**'s light entering surface **190** is marked $\Delta 194$. In a known STC, both rotation axes are located in a center region of OPFE **102**, i.e. a ratio between $\Delta 192$ and L.sub.O and between $\Delta 194$ and H.sub.O respectively is about 0.5. Specifically, $\Delta 192/L.sub.O$ and $\Delta 194/H.sub.O$ in general satisfy $\Delta 192/L.sub.O = 0.3-0.7$ and $\Delta 194/H.sub.O = 0.3-0.7$.

(37) In the following, a “first rotation axis” of a prism indicates the rotation axis that does neither intercept with the light entering surface nor with the light exiting surface of a prism and which is parallel to both the light entering surface nor with the light exiting surface of a prism, as for example first rotation axis **192**. A “second rotation axis” of a prism indicates the rotation axis that intercepts with the light entering surface a prism and which is parallel to the light exiting surface of a prism, as for example second rotation axis **194**. It is noted that the first rotation axis as defined above represents a “fast scan axis” (or “efficient scan axis”) of a STC, as for each degree of rotational movement of a prism around the first rotation axis, s-FOV.sub.T moves by two degrees. The second rotation axis as defined above represents a “slow scan axis” (or “inefficient scan axis”) of a STC, as for each degree of rotational movement of a prism around the second rotation axis, s-FOV.sub.T moves by one degree.

(38) In all examples disclosed herein, the OPFE is a prism having a light entering surface, a light reflecting surface and a light exiting surface. Therefore, we may use “OPFE” and “prism” interchangeably. However, this is not limiting, and in other examples a mirror having a light entering surface may be used.

(39) FIG. 2A shows a STC disclosed herein and numbered **200** in a cross-sectional view. STC **200** includes an OPFE **202** (e.g. a prism or a mirror), a lens **204** including N=6 lens elements L.sub.1-L.sub.6, an (optional) optical filter **205** and an image sensor **206**. FIG. 2B shows STC **200** in a top view.

(40) Lens **204** has an optical axis **208**. STC **200** has an aperture **209**. STC **200** includes a camera module housing **210**. Module housing **210** has a module region **214** having a module height (“H.sub.M”) as well as a module length L.sub.M,1 and a shoulder region **212** having a shoulder height (“H.sub.S”) that is lower by ΔH than module region **214**, i.e. $H_{\text{sub.M}} > H_{\text{sub.S}}$, as well as a shoulder length L.sub.M,2. Here and in the following, all widths (“W”) are measured along an axis parallel to the x-axis, all heights (“H”) are measured along an axis parallel to the y-axis, all lengths (“L”) are measured along an axis parallel to the z-axis.

(41) A theoretical limit for a module height of camera **200** is “minimum module height” (or “MH.sub.M”). A theoretical limit for a shoulder height of camera **200** is “minimum shoulder height” (or “MH.sub.S”). MH.sub.M and MH.sub.S respectively are defined by the largest height dimension of a component included in STC **200**. MH.sub.M is defined by OPFE **202**'s height H.sub.O plus an additional height required for rotating OPFE **202**, as shown. In all STCs disclosed herein, a relatively low MH.sub.M is achieved by making the two following design choices: 1. By locating or positioning a first OPFE rotation axis such as **402**, **452** and **1306** (see FIGS. 13A-13B) relatively close to the light exiting surface of an OPFE, i.e. by relatively small ratios of $\Delta 402/L_{\text{sub.O}}$, $\Delta 452/L_{\text{sub.O}}$ and $\Delta 1306/L_{\text{sub.P}}$ which are smaller than 0.25. 2. Using an OPFE such as OPFE **202**, OPFE **252** and OPFE **1102** that fulfill $H_{\text{sub.O}} < L_{\text{sub.O}}$.

MH.sub.S is defined by image sensor **206**'s height (“H.sub.sensor”) plus an additional height required for rotating OPFE **202**. Small MH.sub.M and MH.sub.S are beneficial for incorporating in slim mobile devices such as smartphones and tablets.

(42) To clarify, all camera modules and optical lens systems disclosed herein are beneficially for use in mobile devices such as smartphones, tablets etc.

(43) For achieving realistic estimations, we calculate H.sub.M and H.sub.S respectively by adding an additional height penalty of 1.5 mm to MH.sub.M or MH.sub.S, i.e. $H_{\text{sub.M}} = MH_{\text{sub.M}} + 1.5$ mm and $H_{\text{sub.S}} = MH_{\text{sub.S}} + 1.5$ mm. The penalty accounts for movements that may be required for optical image stabilization (OIS), autofocus (AF) as well as housing, lens cover etc. Note that the value of 1.5 mm is exemplary and by no means limiting, and that the addition may vary between 1 and 3 mm.

(44) Lens **204** is divided in two lens groups, a first lens group (“G1”) including L1 and L2 and a second lens group (“G2”) including L3-L6. G1 has a maximal optical lens height H.sub.G1 and G2 has a maximal optical lens height H.sub.G2, wherein $H_{\text{sub.G1}} > H_{\text{sub.G2}}$. G1 may be included in the module region **214** and G2 may be included in the shoulder region **212**. G1 has a maximal optical lens width W.sub.G1 and G2 has a maximal optical lens width W.sub.G2, wherein $W_{\text{sub.G1}} > W_{\text{sub.G2}}$ (FIG. 2B). Also OPFE **202** may be included in module region **214**, and optical filter **205** and image sensor **206** may be included in shoulder region **212**. In other embodiments, entire lens **204** (i.e. both G1 and G2) may be included in shoulder region **212**. This may be beneficial for integrating STC **200** in a slim mobile device, i.e. a mobile device having a low height. A large H.sub.G1 may be desired, as it allows STC **200** to have a large DA. A small H.sub.G2 may be desired, as it allows STC **200** to have a slim shoulder region **214**, i.e. small H.sub.S.

(45) For scanning a scanning Tele FOV (“s-FOV.sub.T”) with STC **200**'s native FOV.sub.T (“n-FOV.sub.T”), OPFE **202** is rotated along two dimensions. OPFE **202** is shown in several rotation states which are required for scanning s-FOV.sub.T. The rotation for scanning s-FOV may be actuated by a voice coil motor (VCM). OPFE **202** is a cut (or “D-cut”) prism.

(46) FIG. 2C shows another STC disclosed herein and numbered **250** in a cross-sectional view.

(47) FIG. 2D shows STC **250** in a top view. STC **250** includes an OPFE **252** (e.g. a prism or a

mirror), a lens **254** including $N=6$ lens elements $L_{sub.1}$ - $L_{sub.6}$, an (optional) optical filter **255** and an image sensor **256**. OPFE **252** is shown in several rotation states which are required for scanning s-FOV.sub.T. OPFE **252** is a cut (or “D-cut”) prism.

(48) Lens **254** is a cut lens. Lens **254** has an optical axis **258**, an optical lens height $H_{sub.L}$ and an optical lens width $W_{sub.L}$. STC **250** has an aperture **259**. STC **250** includes a camera module housing **260**. Module housing **260** has a module region **264** with module height $H_{sub.M}$ as well as a module region length $L_{sub.M,1}$ and a shoulder region **262** having shoulder height $H_{sub.S}$ that is lower by ΔH than $H_{sub.M}$, i.e. $H_{sub.M} > H_{sub.S}$, as well as a shoulder region length $L_{sub.M,2}$. For industrial design reasons, it is beneficial to minimize $L_{sub.M,1}$, as it allows for mobile devices with a small $L_{sub.B}$ (FIG. 3B).

(49) A theoretical limit for a module height and a shoulder height of STC **250** is $MH_{sub.M}$ and $MH_{sub.S}$ respectively, as defined above. $H_{sub.M}$ and $H_{sub.S}$ are calculated respectively by adding a penalty 1.5 mm to $MH_{sub.M}$ or $MH_{sub.S}$, i.e. $H_{sub.M} = MH_{sub.M} + 1.5$ mm and $H_{sub.S} = MH_{sub.S} + 1.5$ mm.

(50) Lens **254** is fully included in shoulder region **262**. OPFE **252** is included in module region **264**. Optical filter **255** and image sensor **256** are included in shoulder region **262**.

(51) In other examples, one or more of the first lens elements may be included in module region **264**. For lens **254**, $L_{sub.1}$, which has a larger height $H_{sub.L1}$ than all other lens elements, may be included in module region **264**.

(52) FIG. 3A shows a mobile device **300** (e.g. a smartphone) including STC **200** from FIGS. 2A-B in a cross-sectional view. Mobile device **300** has a front surface **302** (e.g. including a screen, not shown) and a rear surface **310** including STC **200**'s aperture **209**. Mobile device **300** has a regular region **312** of thickness “T” and a camera bump region **314** that is elevated (protrudes outwardly) by a height B over regular region **312**. The bump region has a bump length (“ $L_{sub.B}$ ”) and a bump thickness $T+B$. From an industrial design point of view, it is desired to minimize a bump area, i.e. to have a short $L_{sub.B}$. For achieving short $L_{sub.B}$, module region **214** of STC **200** is included in bump region **314** and shoulder region **212** of STC **200** is included in regular region **312**. This means that OPFE **202** and G1 of lens **204** are included in bump region **314**, and G2 of lens **204** and image sensor **206** are included in regular region **312**.

(53) Optionally, in some embodiments (also referred to as “examples”), parts of shoulder region **212** may also be included in bump region **314**. In other embodiments, both G1 and G2 of lens **204**, i.e. the entire lens **204**, are included in bump region **314**.

(54) FIG. 3B shows a mobile device numbered **320** (e.g. a smartphone) including STC **250** from FIGS. 2C-D in a cross-sectional view. Mobile device **320** has a front surface **322** (e.g. including a screen, not shown) and a rear surface **330** including STC **250**'s aperture **259**. Mobile device **320** has a regular region **332** of thickness “T” and a camera bump region **334** that is elevated by a height B over regular region **332** and has a bump region length $L_{sub.B}$. For achieving short $L_{sub.B}$, module region **264** of STC **250** is included in bump region **334** and shoulder region **262** of STC **250** is included in regular region **332**. OPFE **252** is included in bump region **334** and lens **254** and image sensor **256** are included in regular region **332**. Mobile devices **300** and **320** may additionally include a Wide camera such as Wide camera **130** providing Wide camera images and an application processor (AP). In some examples, the AP may be configured to use Wide camera images to analyze a scene, and, based on the scene analysis, to scan the scene with STC **200**'s and STC **250**'s n-FOV.sub.T autonomously. In other examples, the AP may be configured to scan a scene with STC **200**'s and STC **250**'s n-FOV.sub.T based on a user's input. A FOV of the Wide camera (“FOV.sub.W”) may be in the range of 50 degrees-120 degrees or more, e.g. 80 degrees. In a zero scan position, a center position of n-FOV.sub.T coincides with a center position of FOV.sub.W. A center position of s-FOV.sub.T coincides with a center position of FOV.sub.W. In some examples, s-FOV-covers a 16:9 segment of FOV.sub.W.

(55) FIGS. 3C-E show another mobile device numbered **340** (e.g. a smartphone) including a STC

such as STC **200** from FIGS. 2A-B or STC **250** from FIGS. 2C-D, as well as a Wide camera **344**. In the following and exemplarily, we refer to STC **200** only. FIG. 3C shows mobile device **340** in a perspective view. FIG. 3D shows mobile device **340** in a top view. FIG. 3E shows mobile device **340** in a side view. Wide camera **344** has a Wide camera lens (not shown), a Wide camera aperture **346** and a Wide camera image sensor **348**. Mobile device **340** has a front surface **342** (e.g. including a screen, not shown) and a rear surface **350** including the apertures of the STC and Wide camera **344**. When considered in top view (FIG. 3D), mobile device **340** has a rectangular shape and a device width “W.sub.D” measured along the x-axis as well as a device height “H.sub.D” measured along the z-axis, as shown. A ratio of W.sub.D:H.sub.D is in general different from 1:1, and may be 16:9, 19:9 or similar. That is, in general W.sub.D>H.sub.D. Both image sensor **206** and image sensor **348** have a rectangular shape. In the following, a respective width and height of an image sensor is defined as follows: a width of an image sensor represents (or indicates) a largest dimension of the image sensor, and a height of an image sensor represents a second largest dimension of the image sensor. STC **200**'s image sensor **206** has a Tele sensor width (“W.sub.S,T”) measured along the x-axis and a Tele sensor height (“H.sub.S, +”) measured along the y-axis, as shown. Wide camera **344**'s image sensor **348** has a Wide sensor width (“W.sub.S,W”) measured along the x-axis and a Wide sensor height (“H.sub.S,W”) measured along the z-axis, as shown. In general, Wide camera **344** is integrated into mobile device **340** such that W.sub.S,W is substantially parallel to W.sub.D. For both image sensors, a ratio of W.sub.S:H.sub.S is in general different from 1:1, and may be 4:3, 16:9 or similar. That is, in general W.sub.S>H.sub.S. Image sensor **206** and image sensor **348** may or may not have an identical ratio of W.sub.S:H.sub.S. In mobile device **340**, STC **200** is integrated into mobile device **340** such that W.sub.S, is substantially parallel to both W.sub.S,W and W.sub.D. H.sub.S,T is substantially perpendicular to H.sub.S,W. When considering this incorporation of STC **200** into mobile device **340** in a side view, image sensor **206** is oriented parallel to mobile device **340** (i.e. the rectangular shape of image sensor **206** is oriented parallel to the rectangular shape of mobile device **340**). Therefore, in the following we refer to this configuration as “parallel STC sensor configuration”.

(56) FIGS. 3F-H show yet another mobile device numbered **360** (e.g. a smartphone) including a STC **1250** as well as a Wide camera **344**. STC **1250** includes an optical lens system such as optical lens system **1100** (see FIG. 11) including an OPFE **1102**, a lens (not shown) and an image sensor **1106**. FIG. 3F shows mobile device **360** in a perspective view. FIG. 3G shows mobile device **360** in a top view. FIG. 3H shows mobile device **360** in a side view. Mobile device **360** has a front surface **362** and a rear surface **370** including the apertures of STC **1250** and Wide camera **344**. Mobile device **360** has a device width “W.sub.D” measured along the x-axis as well as a device height “H.sub.D” measured along the z-axis, as shown. Both image sensor **1106** and image sensor **348** have a rectangular shape. STC **1250**'s image sensor **1106** has a Tele sensor width (“W.sub.S,T”) measured along the y-axis and a Tele sensor height (“H.sub.S,T”) measured along the x-axis, as shown. Wide camera **344**'s image sensor **348** has a Wide sensor width (“W.sub.S,W”) measured along the x-axis and a Wide sensor height (“H.sub.S,W”) measured along the z-axis, as shown. In general, Wide camera **344** is integrated into mobile device **340** such that W.sub.S,W is substantially parallel to W.sub.D. For both image sensors, a ratio of W.sub.S:H.sub.S is in general different from 1:1, and may be 4:3, 16:9 or similar. Image sensor **1106** and image sensor **348** may or may not have an identical ratio of W.sub.S:H.sub.S. In mobile device **360**, STC **1250** is integrated into mobile device **360** such that W.sub.S,T is substantially perpendicular to both W.sub.S,W and W.sub.D, which are parallel to each other. H.sub.S,T is substantially perpendicular to H.sub.S,W substantially parallel to W.sub.S,W. When considering this incorporation of STC **1300** into mobile device **360** in a side view, image sensor **1106** is oriented anti-parallel to mobile device **360** (i.e. the rectangular shape of image sensor **1106** is oriented anti-parallel to the rectangular shape of mobile device **360**). Therefore, in the following we refer to this configuration as “anti-parallel STC sensor configuration”.

(57) FIG. 4A shows STC **200** from FIGS. 2A-B without module housing **210** in a zero scan position in a cross-sectional view. The zero scan position is defined by OPFE **202**'s top surface being parallel with the z-axis such that n-FOV.sub.T is located at the center of s-FOV.sub.T. OPFE **202** is rotated around a first rotation axis **402** which is parallel to the x-axis (i.e. perpendicular to both the y-axis and the z-axis shown). Non-cut center axis **404** indicates a center of a non-cut OPFE **202** with respect to the y-axis, i.e. if OPFE **202** was not cut, non-cut center axis **404** would be located at the center of the non-cut OPFE. Cut center axis **406** indicates a center of cut OPFE **202** with respect to the y-axis, i.e. cut center axis **406** is located at the center of cut OPFE **202**. As visible, first rotation axis **402** intersects with the optical axis **208** of lens **204**. However, first rotation axis **402** does not intersect with non-cut center axis **404** nor with cut center axis **406**. First rotation axis **402** is located at distance ΔC from non-cut center axis **404**. This de-center location with respect to the y-axis of OPFE **202** is beneficial for minimizing MH.sub.M. Image sensor **206** is oriented in a parallel STC sensor configuration.

(58) FIG. 4B shows STC **200** without module housing **210** from FIG. 4A in a maximal counter-clockwise rotation state with respect to the first rotation axis **402**. The counter-clockwise rotation direction **412** is shown.

(59) FIG. 4C shows STC **200** without module housing **210** from FIGS. 4A-B in a maximal clockwise rotation state with respect to the first rotation axis **402**. The clockwise rotation direction **414** is shown.

(60) FIG. 4D shows STC **250** from FIGS. 2C-D without module housing **260** in a zero scan position in a cross-sectional view. OPFE **252** is rotated around a first rotation axis **452** which is parallel to the x-axis (i.e. perpendicular to both the y-axis and the z-axis shown). Non-cut center axis **454** indicates a center of a non-cut OPFE **252** with respect to the y-axis. Cut center axis **456** indicates a center of cut OPFE **252** with respect to the y-axis. First rotation axis **452** intersects with the optical axis **258** of lens **254**, but first rotation axis **452** does not intersect with non-cut center axis **454** nor with cut center axis **456**. First rotation axis **452** is located at distance ΔC from non-cut prism center axis **454**. This de-center location of OPFE **252** is beneficial for minimizing MH.sub.M. Image sensor **256** is oriented in a parallel STC sensor configuration.

(61) FIG. 4E shows STC **250** without module housing **260** from FIG. 4D in a maximal counter-clockwise rotation state with respect to the first rotation axis **452**.

(62) FIG. 4F shows STC **250** without module housing **260** from FIGS. 4D-E in a maximal clockwise rotation state with respect to the first rotation axis **452**.

(63) The counter-clockwise rotation direction **462** and the clockwise rotation direction **464** are shown.

(64) FIG. 5A shows STC **200** from FIGS. 2A-B without module housing **210** in a zero scan position in a top view. The zero scan position is defined by OPFE **202**'s top surface being parallel with the z-axis such that n-FOV.sub.T is located at the center of s-FOV.sub.T. OPFE **202** is rotated around a second rotation axis **502** which is parallel to the y-axis (i.e. perpendicular to both the x-axis and the z-axis shown).

(65) FIG. 5B shows STC **200** without module housing **210** from FIG. 5A in a maximal clockwise rotation state with respect to the second rotation axis **502**. The clockwise rotation direction **512** is shown.

(66) FIG. 5C shows STC **200** without module housing **210** from FIGS. 5A-B in a maximal counter-clockwise rotation state with respect to the second rotation axis **502**. The counter-clockwise rotation direction **514** is shown.

(67) FIG. 5D shows STC **250** from FIGS. 2C-D without module housing **260** in a zero scan position in a top view. OPFE **252** is rotated around a second rotation axis **552** which is parallel to the y-axis (i.e. perpendicular to both the x-axis and the z-axis shown).

(68) FIG. 5E shows STC **250** without module housing **260** from FIG. 5D in a maximal clockwise rotation state with respect to the second rotation axis **552**.

(69) FIG. 5F shows STC 250 without module housing 210 from FIGS. 5D-E in a maximal counter-clockwise rotation state with respect to the second rotation axis 552.

(70) The clockwise rotation direction 562 and the counter-clockwise rotation direction 564 are shown. FIG. 6A shows OPFE 202 of STC 200 in a zero scan position in a cross-sectional view. The location of first rotation axis 402 and second rotation axis 502 are shown. A distance from first rotation axis 402 to OPFE 202's light exiting surface 604 is marked $\Delta 402$. Here, $\Delta 402=0.5$ mm. OPFE 202 is a cut prism which has a cut bottom corner (or margin) 612. Considering the cutting along the z-axis, a cut surface extends from light exiting surface 604 along a distance Δ_{cut} , as shown. Beneficially, first rotation axis 402 is located within this distance Δ_{cut} . This is valid also for all other OPFEs disclosed herein such as OPFE 202, OPFE 252 and OPFE 1102.

(71) A distance from second rotation axis 502 to OPFE 202's light entering surface 602 is marked $\Delta 502$. Here, $\Delta 502=4.3$ mm.

(72) FIG. 6B shows OPFE 202 of STC 200 from FIG. 6A in a cross-sectional view. A first rotation direction 606 around first rotation axis 402 and a second rotation direction 608 around second rotation axis 404 are shown. A prism height H.sub.P and a prism length L.sub.P OPFE 202 are shown as well. Here, L.sub.P=7.1 mm and H.sub.P=6.9 mm. Ratios of $\Delta 402/L_{\text{sub.P}}=0.07$ and $\Delta 502/H_{\text{sub.P}}=0.62$.

(73) FIG. 6C shows OPFE 252 of STC 250 in a zero scan position in a cross-sectional view. OPFE 252 is a prism. The location of first rotation axis 452 and second rotation axis 552 are shown. A distance from first rotation axis 452 to OPFE 252's light exiting surface 654 is $\Delta 452$. Here, $\Delta 452=0.5$ mm. A distance from second rotation axis 552 to OPFE 252's light entering surface 652 is $\Delta 552$. Here, $\Delta 552=3.5$ mm.

(74) FIG. 6D shows OPFE 252 of STC 250 from FIG. 6C in a cross-sectional view. A first rotation direction 656 around first rotation axis 452 and a second rotation direction 658 around second rotation axis 454 as well as a length L.sub.P and height H.sub.P of OPFE 252 are shown. Here, L.sub.P=7.2 mm and H.sub.P=6.7 mm. Ratios of $\Delta 452/L_{\text{sub.P}}=0.07$ and $\Delta 552/H_{\text{sub.P}}=0.52$.

(75) FIG. 7 shows exemplary a Wide camera FOV ("FOV.sub.W"), a s-FOV.sub.T and 9 n-FOV.sub.Ts (marked 1-9) of a known Wide camera and a STC such as STC 200 or STC 250 or STC 1250. FOV.sub.W shows a typical Wide camera FOV of for example 82° measured along a diagonal of FOV.sub.W. In this example, a 16:9 FOV ratio of FOV.sub.W covers about $69.4^\circ \times 42.6^\circ$ (i.e. 69° in a horizontal direction, 42° in a vertical direction). s-FOV-shows a segment of a scene that can be covered with the STC, i.e. it includes all POVs that can be reached with the STC. In some examples, s-FOV.sub.T may cover a 16:9 FOV ratio of FOV.sub.W, as shown in FIG. 7. As visible, FOV.sub.W and s-FOV.sub.T have a "longer side" 702 (here, along the z-axis) and a "shorter side" 704 (here, along the z-axis). s-FOV.sub.T of STC 200 covers $50.9^\circ \times 32.5^\circ$ (50.9° in a horizontal direction, 32.5° in a vertical direction). The 9 n-FOV.sub.Ts represent maximum scan positions. n-FOV.sub.T 5, i.e. the (Center, Center) position, represents a zero scan position. For example, n-FOV.sub.T 1 represents the n-FOV.sub.T that is obtained when scanning STC 200 maximally to a top-left position, n-FOV.sub.T 6 represents the n-FOV.sub.T that is obtained when scanning STC 200 maximally to a bottom-center position etc. Table 1 provides the rotation values of OPFE 202 around (first rotation axis 402, second rotation axis 502) respectively that are required for scanning to the 9 respective n-FOV.sub.Ts. The values refer to a scanning action that starts from n-FOV.sub.T 5, i.e. the (Center, Center) position. For example for scanning n-FOV.sub.T to n-FOV.sub.T 9 or (Bottom, Right), starting from (Center, Center) position n-FOV.sub.T 5, OPFE 202 must be rotated by -7.85 degrees around first rotation axis 402 and by -15.46 degrees around second rotation axis 502.

(76) TABLE-US-00001 TABLE 1 Left Center Right Top (1.76, 21.67) (4.86, 0) (1.76, -21.67) Center (-3.16, 18.49) (0, 0) (-3.16, -18.49) Bottom (-7.85, 15.46) (-4.86, 0) (-7.85, -15.46)

(77) For STC 250 including optical lens system 900, Table 2 provides the rotation values of OPFE 252 around (first rotation axis 452, second rotation axis 552) respectively that are required for

scanning to the 9 respective n-FOV.sub.Ts shown in FIG. 7. s-FOV.sub.T of STC **250** covers 69.5°×42.58°. This means that s-FOV.sub.T of STC **250** covers a 16:9 ratio of a FOV.sub.W having a diagonal FOV.sub.W=82°, as shown in FIG. 7

(78) TABLE-US-00002 TABLE 2 Left Center Right Top (3.63, 29.80) (8.35, 0) (3.63, -29.80) Center (-5.80, 24.02) (0, 0) (-5.80, -24.02) Bottom (-11.67, 18.83) (-8.35, 0) (-11.67, -18.83)

(79) For another STC (not shown) including optical lens system **1000**, Table 3 provides the rotation values of OPFE **1002** around a first rotation axis and around a second rotation axis respectively that are required for scanning to the 9 respective n-FOV.sub.Ts shown in FIG. 7. s-FOV of the STC including optical lens system **1000** covers 69.5°×42.58°. This means that the s-FOV-covers a 16:9 ratio of a FOV.sub.W having a diagonal FOV.sub.W=82°.

(80) TABLE-US-00003 TABLE 3 Left Center Right Top (1.49, 25.54) (5.79, 0) (1.49, -25.54) Center (-4.31, 21.34) (0, 0) (-4.31, -21.34) Bottom (-9.67, 17.16) (-5.79, 0) (-9.67, -17.16)

In some examples, an OPFE may be rotated around one axis or around two axes for optical image stabilization (OIS). In some examples and per axis, an OPFE may be rotated by ±2 degrees or by ±5 degrees around a zero position for performing OIS. In other examples, an OPFE may be rotated by even ±10 degrees or more around a zero position for performing OIS. In these examples, in general a mobile device including the STC includes as well an additional sensor such as e.g. an inertial measurement unit (IMU) and a processor, e.g. an application processor (AP) or a micro controller unit (MCU). The additional sensor is used for sensing an undesired rotation of the mobile device, and based on the sensing data of the additional sensor, the processor calculates OPFE rotation control signals which control a rotational movement of the OPFE that mitigates (or counteracts) the undesired rotation of the mobile device.

(81) FIG. 8 shows optical lens system **800** included in STC **200** from FIGS. 2A-B in a cross-sectional view and with ray-tracing. A distance ΔL.sub.O between OPFE **202** and lens **204** is 2.7 mm.

(82) The optical height (H.sub.L1) and width (W.sub.L1) of lens element L.sub.1 may define the optical height and width of G1 (i.e. H.sub.L1=H.sub.G1 and W.sub.L1=W.sub.G1) as well as an aperture of camera **200**, such that the optical height and the optical width of lens element L.sub.1 represent also the aperture height (HA) and aperture width (WA) of lens **204** respectively. The D-cut of L1 and G1 means that also STC **200**'s aperture changes accordingly, such that the aperture is not axial symmetric. The cutting allows for a small lens heights H.sub.G1, which are required for small MH.sub.Ms, and still relatively large effective aperture diameters (DAs) which satisfy DA>H.sub.G1.

(83) In other examples, an EFL of lens **204** may be 8 mm-50 mm.

(84) G2 is D-cut as well. The optical height (H.sub.L3) and width (W.sub.L3) of lens element L.sub.3 may define the optical height, width and aperture of G2. Prism **202** is D-cut as well.

(85) Detailed optical data and surface data are given in Tables 2-3 for the example of the lens elements in FIG. 8. The values provided for these examples are purely illustrative and according to other examples, other values can be used.

(86) Surface types are defined in Table 4. The coefficients for the surfaces are defined in Table 5. The surface types are: a) Plano: flat surfaces, no curvature b) Q type 1 (QT1) surface sag formula:

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2 r^2}} + D_{con}(u)$$

$$(87) \quad D_{con}(u) = u^4 \cdot \text{Math} \cdot \prod_{n=0}^N A_n Q_n^{con}(u^2) \quad (\text{Eq. 1})$$

$$u = \frac{r}{r_{norm}}, x = u^2$$

$$Q_{sub.0.sup.con}(x)=1 \quad Q_{sub.1.sup.con}=-(5-6x) \quad Q_{sub.2.sup.con}=15-14x(3-2x)$$

$$Q_{sub.3.sup.con}=-\{35-12x[14-x(21-10x)]\}$$

Q.sub.4.sup.con=70-3x{168-5x[84-11x(8-3x)]}

Q.sub.5.sup.con=-[126-x(1260-11x{420-x[720-13x(45-14x)]})] c) Even Asphere (ASP) surface sag formula:

(88)

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2 r^2}} + {}_1r^2 + {}_2r^4 + {}_3r^6 + {}_4r^8 + {}_5r^{10} + {}_6r^{12} + {}_7r^{14} + {}_8r^{16} \quad (\text{Eq. 2})$$

(89) TABLE-US-00004 Example 800 EFL = 17.37 mm, f number = 2.35 (Eff. DA/2 = 3.7 mm), HFOV = 12.8 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2) Material Index Abbe # Length 1 A.S. Plano Infinity -1.543 4.000 2 Lens 1 ASP 5.876 3.001 4.000 Glass 1.48 84.1 10.424 3 -30.470 0.373 3.666 4 Lens 2 ASP -34.332 2.433 3.515 Plastic 1.61 25.6 -14.027 5 11.925 0.892 2.892 6 Lens 3 ASP -30.326 2.043 2.774 Plastic 1.67 19.2 14.887 7 -7.770 0.035 2.782 8 Lens 4 ASP -22.896 1.591 2.672 Plastic 1.54 55.9 24.980 9 -8.758 0.224 2.781 10 Lens 5 ASP -69.824 1.017 2.618 Plastic 1.61 25.6 -110.272 11 2962.171 0.927 2.516 12 Lens 6 ASP -4.642 0.347 2.497 Plastic 1.61 25.6 -8.581 13 -38.037 5.279 2.732 14 Filter Plano Infinity 0.210 — Glass 1.52 64.2 15 Infinity 0.350 — 16 Image Plano Infinity — —

(90) where {z, r} are the standard cylindrical polar coordinates, c is the paraxial curvature of the surface, k is the conic parameter, r.sub.norm is generally one half of the surface's clear aperture, and A.sub.n are the polynomial coefficients shown in lens data tables. The Z axis is positive towards the image. Values for aperture radius are given as a clear aperture radius, i.e. DA/2. The reference wavelength is 555.0 nm. Units are in mm except for refraction index ("Index") and Abbe #.

(91) The same formulas, units and definitions are used also for Tables 6-11.

(92) TABLE-US-00005 TABLE 5 Aspheric Coefficients Surface # Conic 4.sup.th 6.sup.th 8.sup.th 10.sup.th 2 0 -2.45E-06 1.03E-05 -9.26E-07 8.22E-08 3 0 -7.38E-05 1.77E-06 -1.09E-06 1.02E-07 4 0 -1.19E-03 -1.87E-05 1.97E-06 1.94E-07 5 0 -2.78E-04 -2.26E-04 -8.07E-06 2.41E-06 6 0 1.31E-03 -5.27E-04 7.78E-06 -2.89E-06 7 0 5.97E-04 -2.77E-04 -1.68E-05 5.15E-06 8 0 -2.20E-03 1.74E-04 -3.81E-05 1.19E-06 9 0 -1.95E-03 -1.73E-04 -6.31E-06 3.10E-07 10 0 1.16E-04 -8.54E-04 3.38E-05 8.06E-06 11 0 4.28E-04 -7.22E-04 4.63E-05 3.53E-06 12 0 -3.11E-03 -1.03E-04 -6.08E-06 3.71E-07 13 0 -3.09E-03 1.09E-04 8.66E-06 -2.01E-06 Aspheric Coefficients (Continued) Surface # 12.sup.th 14.sup.th 16.sup.th 2 -2.68E-09 5.28E-13 -2.97E-14 3 -1.97E-09 -7.40E-14 9.91E-15 4 -7.91E-09 3.55E-12 4.10E-13 5 -8.91E-08 1.09E-09 8.66E-13 6 5.28E-07 -1.95E-08 -4.33E-13 7 -2.86E-07 7.01E-09 6.35E-12 8 1.26E-07 -6.82E-09 -5.07E-12 9 2.95E-07 -3.40E-08 1.09E-09 10 -1.52E-07 -2.21E-08 1.29E-10 11 2.90E-08 -2.34E-08 6.33E-10 12 2.55E-08 -6.72E-08 5.93E-09 13 -1.59E-07 3.54E-08 -1.12E-09

(93) FIG. 9 shows optical lens system 900 included in STC 250 from FIGS. 2C-D in a cross-sectional view and with ray-tracing. ΔL.sub.O is 2.7 mm, ΔC is 0.15 mm.

(94) The optical height (H.sub.L1) and width (W.sub.L1) of lens element L.sub.1 may define the optical height and width of lens 254 as well as an aperture of STC 250, such that the optical height and the optical width of lens element L.sub.1 represent also the aperture height (HA) and aperture width (WA) of lens 254 respectively. The D-cut of L.sub.1 means that also STC 250's aperture changes accordingly. The cutting allows for a small HA and still relatively large effective DAs which satisfy DA>HA. In other examples, an EFL of lens 254 may be 8 mm-50 mm. Prism 252 is D-cut as well. A s-FOV.sub.T is 69.5deg×42.58deg, i.e. a horizontal direction of s-FOV.sub.T ("H-s-FOV.sub.T") is H-s-FOV.sub.T=69.5deg, a vertical direction of s-FOV.sub.T ("V-s-FOV.sub.T") is V-s-FOV.sub.T=42.58deg. s-FOV.sub.T covers a 16:9 FOV ratio of a FOV.sub.W=82deg (diagonal) of a Wide camera that may be included in a mobile device together with the STC.

(95) Detailed optical data and surface data are given in Tables 6-7.

(96) TABLE-US-00006 TABLES 6-7 Aspheric Coefficients Surface # Conic 4.sup.th 6.sup.th

8.sup.th 10.sup.th 2 0 -3.85E-04 -1.13E-05 -2.76E-06 -9.98E-08 3 0 8.43E-04 -5.11E-05
-4.46E-07 2.27E-07 4 0 -1.17E-03 1.34E-06 -2.87E-06 1.51E-06 5 0 2.58E-03 1.86E-05
6.27E-05 8.43E-06 6 0 2.92E-03 -3.50E-04 5.10E-05 7.31E-06 7 0 9.39E-04 -1.77E-04
-2.83E-05 3.52E-06 8 0 1.43E-03 -6.10E-04 -1.44E-04 -1.32E-05 9 0 -3.91E-03
-4.05E-04 1.55E-05 -1.41E-06 10 0 -2.93E-03 -5.10E-04 2.28E-04 3.56E-05 11 0
5.67E-03 -4.18E-04 9.28E-05 1.35E-05 12 0 -3.50E-03 -7.08E-04 5.45E-04 -2.00E-04 13
0 -1.21E-02 9.04E-04 -9.24E-05 -7.71E-06 Aspheric Coefficients (Continued) Surface #

12.sup.th 14.sup.th 16.sup.th 2 -5.82E-10 7.38E-10 -1.79E-10 3 -8.02E-09 -1.13E-09
6.18E-11 4 1.92E-07 -6.49E-09 -1.47E-09 5 2.83E-07 1.11E-07 -2.77E-08 6 1.28E-06
-1.04E-07 -1.23E-08 7 8.01E-07 5.64E-07 -6.36E-08 8 -2.77E-06 -5.16E-07 9.10E-08 9
-1.75E-06 -4.88E-07 5.63E-08 10 9.70E-07 -9.49E-07 4.30E-08 11 -9.57E-07 9.48E-07
-1.14E-07 12 -1.59E-05 7.95E-06 -4.71E-07 13 -5.10E-06 9.72E-07 -3.87E-08

(97) TABLE-US-00007 Example 900 EFL = 14.1 mm, f number = 2.45 (Eff. DA/2 = 2.9 mm),
HFOV = 11.5 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness
(D/2) Material Index Abbe # Length 1 A.S. Plano Infinity -1.246 3.000 2 Lens 1 ASP 3.981 1.941
3.013 Glass 1.48 84.1 7.619 3 -44.268 1.288 2.903 4 Lens 2 ASP 23.035 0.281 2.448 Plastic 1.61
25.6 -8.410 5 4.224 0.020 2.232 6 Lens 3 ASP 3.832 0.531 2.251 Plastic 1.53 55.7 32.268 7 4.684
1.504 2.199 8 Lens 4 ASP 43.147 0.869 2.230 Plastic 1.66 20.4 7.554 9 -5.655 0.018 2.326 10
Lens 5 ASP -9.456 0.238 2.231 Plastic 1.61 25.6 -8.644 11 12.423 3.160 2.100 12 Lens 6 ASP
-3.038 1.058 2.193 Plastic 1.53 55.7 -14.591 13 -5.570 0.941 2.654 14 Filter Plano Infinity 0.210
— Glass 1.52 64.2 15 Infinity 0.350 — 16 Image Plano Infinity — —

(98) FIG. 10 shows another optical lens system 1000 disclosed herein in a cross-sectional view and
with ray-tracing. Optical lens system 1000 may be included in a STC such as STC 200 or STC 250.
Optical lens system 1000 includes a prism 1002, a lens 1004 including N=6 lens elements, an
(optional) optical filter 1005 and an image sensor 1006. $\Delta L_{sub.O}$ is 2.7 mm, ΔC is 0.15 mm. A
distance from a first rotation axis to OPFE 1002's light exiting surface is 0.5 mm. A distance from a
second rotation axis to OPFE 1000's light exiting surface is 4.3 mm.

(99) H.sub.L1 and W.sub.L1 of lens element L.sub.1 may define the optical height and width of
lens 1004 as well as an aperture of a STC that includes optical lens system 1000, such that the
optical height and the optical width of lens element L.sub.1 represent also the aperture height (HA)
and aperture width (WA) of lens 1004 respectively. Lens 1004, i.e. L.sub.1 and further lens
elements, as well as prism 1002 are D-cut. In other examples, an EFL of lens 1004 may be 8 mm-
50 mm and SD may be 4 mm-15 mm. Detailed optical data and surface data are given in Tables 8-
9.

(100) TABLE-US-00008 TABLE 8 Example 1000 EFL = 14.1 mm, f number = 2.45 (Eff. DA/2 =
2.9 mm), HFOV = 15.7 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius
Thickness (D/2) Material Index Abbe # Length 1 A.S. Plano Infinity -1.121 3.000 2 Lens 1 ASP
4.321 2.708 3.018 Glass 1.48 84.1 9.046 3 208.561 0.539 2.844 4 Lens 2 ASP 23.819 0.709 2.712
Plastic 1.57 37.4 13.890 5 -11.728 0.062 2.688 6 Lens 3 ASP -8.405 0.308 2.646 Plastic 1.61 25.6
-6.440 7 7.678 0.811 2.474 8 Lens 4 ASP -58.073 1.934 2.505 Plastic 1.67 19.2 8.073 9 -5.062
0.038 2.826 10 Lens 5 ASP -35.454 1.794 2.765 Plastic 1.61 25.6 -11.971 11 9.540 1.157 2.577 12
Lens 6 ASP -67.401 0.525 2.551 Plastic 1.67 19.2 -15.109 13 12.096 3.371 2.909 14 Filter Plano
Infinity 0.210 — Glass 1.52 64.2 15 Infinity 0.350 — 16 Image Plano Infinity — —

(101) TABLE-US-00009 TABLE 9 Aspheric Coefficients Surface # Conic 4.sup.th 6.sup.th
8.sup.th 10.sup.th 2 0 -1.30E-04 -1.20E-05 -1.85E-06 1.87E-07 3 0 7.73E-04 -2.67E-04
-5.52E-06 2.38E-06 4 0 -7.46E-04 -4.36E-04 -1.63E-05 5.83E-06 5 0 -1.90E-03
2.66E-05 -1.30E-06 1.60E-06 6 0 1.67E-03 -5.80E-04 8.83E-05 -5.04E-06 7 0 3.06E-03
-2.99E-04 -5.92E-05 3.02E-05 8 0 -1.90E-03 3.91E-04 -1.26E-04 5.48E-06 9 0
-1.56E-03 2.81E-04 -1.86E-05 -6.04E-06 10 0 -3.81E-03 -1.27E-04 1.13E-04 -1.19E-05
11 0 -3.87E-03 -7.18E-04 1.83E-04 -3.44E-05 12 0 -2.57E-02 1.22E-03 -4.89E-05

-4.84E-06 13 0 -2.37E-02 2.34E-03 -1.49E-04 -1.79E-06 Aspheric Coefficients (Continued)
 Surface # 12.sup.th 14.sup.th 16.sup.th 2 -5.47E-08 3.88E-09 -2.08E-10 3 -1.56E-07
 8.60E-09 -2.61E-10 4 2.11E-08 -4.48E-08 1.98E-09 5 -9.67E-07 9.65E-08 -2.76E-09 6
 5.46E-07 -1.04E-07 4.93E-09 7 -3.15E-06 3.96E-07 -2.26E-08 8 1.13E-06 -1.54E-07
 5.51E-09 9 1.08E-06 -7.98E-08 2.33E-09 10 2.81E-07 1.09E-07 -6.13E-09 11 4.58E-06
 -4.02E-07 1.43E-08 12 -2.96E-06 6.02E-07 -2.54E-08 13 1.11E-06 -5.00E-08 4.36E-10
 (102) FIG. 11 shows another optical lens system 1100 disclosed herein with ray-tracing. FIG. 11A shows optical lens system 1100 in a cross-sectional view. FIG. 11B shows optical lens system 1100 in a top view. FIG. 11C shows optical lens system 1100 in a perspective view.

(103) Optical lens system 1100 includes an OPFE 1102 (e.g. a prism or a mirror), a lens 1104 including N=6 lens elements L.sub.1-L.sub.6, an (optional) optical filter 1105 and an image sensor 1106. Lens 1104 has an optical axis 1108. Lens 1104 is a cut lens. The cutting is performed such that a height of lens 1104 ("H.sub.L", measured along the y-axis) is 5.1 mm, as shown in FIG. 11A. This means that a cut ratio, i.e. a ratio between which a height of a lens element differs from its width, is 20% or less. Cutting a lens like lens 1104 is beneficial as of two desired outcomes: it reduces a height of the cut lens itself (what reduces MH.sub.S) and it reduces a height of an OPFE such as OPFE 1102 (what reduces MH.sub.M). Specifically, cutting a lens by X % will reduce MH.sub.M and MH.sub.S by about 0.5.Math.X %-X %. For example, cutting a lens by 20% will reduce MH.sub.M and MH.sub.S by about 10%-20%.

(104) In FIG. 11C is visible that image sensor 1106 is oriented in an anti-parallel STC sensor configuration. Orienting image sensor 1106 in an anti-parallel STC sensor configuration is beneficial as it aligns a fast scan axis of OPFE 1102 with a longer side of s-FOV.sub.T and a slow scan axis of OPFE 1102 with a shorter side of s-FOV.sub.T. Therefore, a maximum rotational movement required for covering s-FOV.sub.T can be smaller than in STCs that use a parallel STC sensor configuration. For example and for covering a 16:9 ratio of FOV.sub.W=82° (diagonal), for STC 1250 (anti-parallel STC sensor configuration) a maximum rotational movement of OPFE 1102 is 21.35° (see Table 12), whereas for STC 250 and optical lens system 1000 (parallel STC sensor configuration) a maximum rotational movement of OPFE 252 and OPFE 1002 is 29.8° (see Table 2) and 25.5° (see Table 3) respectively. Smaller maximum rotational movements are beneficial, as they can be provided by simpler actuators as well as simpler and more accurate actuation control.

(105) Detailed optical data and surface data are given in Tables 10-11. An effective f/#based on an effective lens aperture diameter as known in the art is given.

(106) TABLE-US-00010 TABLE 10 Example 1100 EFL = 14.1 mm, Eff. f number = 2.43 (Eff. DA/2 = 2.9 mm), HFOV = 13.7 deg. Aperture Curvature Radius Focal Surface # Comment Type
 Radius Thickness (D/2) Material Index Abbe # Length 1 A.S. Plano Infinity -1.058 3.000 2 Lens 1
 ASP 4.430 1.600 3.001 Glass 1.48 84.1 9.556 3 85.664 1.088 2.967 4 Lens 2 ASP 8.627 0.819
 2.773 Plastic 1.53 55.69 12.034 5 -24.809 0.109 2.716 6 Lens 3 ASP -8.650 0.503 2.691 Plastic
 1.61 25.59 -4.544 7 4.256 0.517 2.414 8 Lens 4 ASP 7.547 1.600 2.412 Plastic 1.67 19.24 8.497 9
 -22.121 0.037 2.520 10 Lens 5 ASP 5.421 0.788 2.424 Plastic 1.61 25.59 -15.577 11 3.276 2.994
 2.293 12 Lens 6 ASP -17.791 0.683 2.797 Plastic 1.53 55.69 -81.994 13 -30.272 2.382 3.001 14
 IR Filter Plano Infinity 0.210 — Glass 1.52 64.17 15 Infinity 0.350 — 16 Image Plano Infinity —

(107) TABLE-US-00011 TABLE 11 Aspheric Coefficients Surface # Conic 4.sup.th 6.sup.th
 8.sup.th 10.sup.th 12.sup.th 14.sup.th 16.sup.th 2 0 -5.19E-04 -1.78E-05 -4.48E-06 -8.07E-08
 -1.64E-08 -8.77E-10 1.34E-10 3 0 -1.12E-03 2.33E-04 -4.20E-05 4.57E-06 -4.19E-07
 2.34E-08 -6.24E-10 4 0 -6.44E-03 6.51E-04 -4.40E-05 2.04E-06 8.43E-08 7.98E-09
 -1.44E-09 5 0 -2.21E-04 4.15E-04 3.01E-05 1.90E-05 -3.57E-06 1.45E-07 4.15E-09 6
 0 7.41E-03 -1.55E-03 2.67E-04 -2.54E-05 2.21E-06 -2.67E-07 1.70E-08 7 0 -5.36E-03
 1.97E-04 -3.28E-05 -1.36E-06 1.91E-06 -4.06E-07 4.19E-08 8 0 1.57E-03 -1.49E-03
 2.62E-04 -8.49E-05 1.59E-05 -2.24E-06 1.37E-07 9 0 -8.09E-03 5.64E-04 -3.26E-04

7.59E-05 -9.18E-06 5.63E-07 -1.18E-08 10 0 -3.63E-02 2.08E-03 2.51E-04 -2.96E-05
2.33E-06 -2.20E-07 8.14E-09 11 0 -2.75E-02 1.60E-03 9.49E-04 -3.33E-04 5.92E-05
-5.44E-06 2.06E-07 12 0 -1.02E-02 2.26E-04 -1.51E-04 5.00E-05 -8.69E-06 8.68E-07
-3.28E-08 13 0 -9.62E-03 -1.91E-05 1.57E-05 -3.19E-06 4.16E-07 -3.72E-08 2.12E-09

(108) With reference to FIG. 7, a s-FOV.sub.T of STC **1250** covers $69.4^\circ \times 42.6^\circ$. It is noted that s-FOV.sub.T covers a 16:9 ratio of FOV.sub.W of a Wide (or Main) camera such as **130** having a (diagonal) FOV.sub.W= 82° . Table 12 provides the rotation values of OPFE **1202** around a first and a second rotation respectively that are required for scanning to the 9 respective n-FOV.sub.Ts shown in FIG. 7. The values refer to a scanning action that starts from n-FOV.sub.T 5, i.e. the (Center, Center) position. For example for scanning n-FOV.sub.T to n-FOV.sub.T 9 or (Bottom, Right), starting from (Center, Center) position n-FOV.sub.T 5, OPFE **1202** must be rotated by -13.18 degrees around the first rotation axis and by -8.9 degrees around the second rotation axis.

(109) TABLE-US-00012

TABLE 12	Left	Center	Right	Top	(9.2, 21.35)	(12.05, 0)	(9.2, -21.35)
Center	(-1.99, 17)	(0, 0)	(-1.99, -17)	Bottom	(-13.18, 8.9)	(-12.05, 0)	(-13.18, -8.9)

(110) FIG. 12 shows a mobile device **1200** (e.g. a smartphone) including a STC **1250** in a cross-sectional view. STC **1250** may include optical lens system **1100** shown in FIGS. 11A-C. Mobile device **1200** has a front surface **1202** (e.g. including a screen, not shown) and a rear surface **1210** including STC **1250**'s aperture **1259**. Mobile device **1200** has a regular region **1212** of thickness "T" and a camera bump region **1214** that is elevated by a height B over regular region **1212**. Bump region **1214** has a bump length ("L.sub.B") and a bump thickness T+B. For achieving short L.sub.B, a module region **1252** of STC **1250** having a height MH.sub.M is included in bump region **1214** and shoulder region **1254** of STC **1250** having a height MH.sub.S < MH.sub.M is included in regular region **1312**. This means that OPFE **1102** is included in bump region **1214** and lens **1104** and image sensor **1106** are included in regular region **1212**. Optionally, in some embodiments, parts of shoulder region **1254** may also be included in bump region **1214**.

(111) FIGS. 13A-B show OPFE **1102** (here, a prism) of STC **1250** in a zero scan position. FIG. 13A shows OPFE **1102** in a side (or cross-sectional) view. FIG. 13B shows OPFE **1102** in a perspective view.

(112) OPFE **1102** has a light entering surface **1302** and a light exiting surface **1304**. The location of first rotation axis **1306** and second rotation axis **1312** are shown. OPFE **1102** has a prism height ("H.sub.P") and an optical (or optically active) prism height ("H.sub.P-O"), a prism length ("L.sub.P") and an optical prism length ("L.sub.P-O") and a prism width ("W.sub.P"), as shown.

(113) A distance from first rotation axis **1306** to OPFE **1102**'s light exiting surface **1304** is $\Delta 1306$. Here, $\Delta 1306 = 0.5$ mm and a ratio of $\Delta 1306$ and the prism length L.sub.P is $\Delta 1306 / L_{sub.P} = 0.07$. This de-center location of OPFE **1102** is beneficial for minimizing MH.sub.M. A distance from second rotation axis **1312** to OPFE **1102**'s light entering surface **1302** is $\Delta 1312$. Here, $\Delta 1312 = 3.35$ mm and a ratio of $\Delta 1312$ and the prism height is $\Delta 1312 / H_{sub.P} = 0.55$.

(114) OPFE **1102** has a non-cut center axis **1332** that indicates a center of a non-cut OPFE **1102** with respect to the y-axis. OPFE **1102** has a cut center axis **1334** that indicates a center of cut OPFE **1102** with respect to the y-axis. Both first rotation axis **1306** and second rotation axis **1312** intersect with optical axis **1108** of lens **1104** and with non-cut center axis **1332**. In other words and referring to FIG. 4D and FIGS. 6A-D, in optical lens system **1100** $\Delta C = 0$.

(115) OPFE **1102** includes an exiting-surface top stray light prevention mask **1322** having a height H.sub.T-SM, an exiting-surface bottom stray light prevention mask **1324** having a height H.sub.B-SM, an entering-surface left stray light prevention mask **1326** having a length LL-SM and an entering-surface right stray light prevention mask **1328** having a length L.sub.R-SM. Values and ranges are given in Table 13 in mm. The stray light prevention masks are beneficial because they prevent stray light from reaching an image sensor such as image sensor **1106**. Stray light is undesired light emitted or reflected from an object in a scene which enters a camera's aperture and

reaches an image sensor at a light path that is different from a planned (or desired) light path. A planned light path is described as follows: 1. Light is emitted or reflected by an object in a scene. 2. Light enters a camera's aperture. 3. For examples where the OPFE is a mirror, light is reflected once at the mirror's surface. For examples where the OPFE is a prism, light passes once a light entering surface of the prism, is reflected once at the prism's reflective surface, and then passes once a light exiting surface of the prism. 4. Light passes once all surfaces of a lens. 5. Light impinges on an image sensor.

Light that reaches an image sensor on any light path other than the planned light path described above is considered undesired and referred to as stray light.

Values and ranges are given in Table 13 in mm.

(116) L.sub.P-O/L.sub.P=0.76, i.e. left stray light prevention mask **1326** and right stray light prevention mask **1328**, which are located at the light entering surface **1302**, together cover a surface area of more than 20% and less than 30% of the area of the light entering surface **1302**. H.sub.P-O/H.sub.P=0.83, i.e. top stray light prevention mask **1322** and bottom stray light prevention mask **1324** which are located at the light exiting surface **1304**, together cover a surface area of more than 10% and less than 20% of the area of the light entering surface **1304**.
 (117) TABLE-US-00013 TABLE 13 Value Value range H.sub.P 6.11 3-10 H.sub.P-O 5.1 2-10 L.sub.P 6.72 3-12 L.sub.P-O 5.13 2-12 W.sub.P 10.3 4-15 H.sub.T-SM 0.81 0.1-2.5 H.sub.B-SM 0.2 0.05-2.5 L.sub.L-SM 0.83 0.1-4 L.sub.R-SM 0.76 0.1-4 Δ 1306 0.5 0.2-3 Δ 1312 3.35 1.5-6

Table 14 summarizes values and ratios thereof of various features that are included in STC **200**, STC **250** and STC **1230** and optical lens systems **800**, **900**, **1000** and **1100**. H.sub.G1, W.sub.G1, H.sub.G2, W.sub.G2, Δ C, HA, WA, DA, HA.sub.G2, WA.sub.G2, DA.sub.G2, H.sub.P, W.sub.P, L.sub.P, Δ LO, TTL, BFL, EFL, EFL.sub.G1, EFL.sub.G2, SD, H.sub.sensor, MH.sub.S, MH.sub.M, H.sub.S, H.sub.M, ALT, ALT.sub.G1, ALT.sub.G2, T.sub.1, f.sub.1 are given in mm. n-FOV.sub.T, S-FOV.sub.T, α -OPFE and β -OPFE are given in degrees.

(118) In other examples, the values may differ from the values given here by e.g. $\pm 10\%$, or by $\pm 20\%$, or by even $\pm 30\%$. "Type" specifies whether the optical lens system uses a parallel STC sensor configuration ("P") or an anti-parallel STC sensor configuration ("A-P"). "16:9 W ratio" indicates whether the s-FOV.sub.T of the respective optical lens system covers ("Y") or not covers ("N") a 16:9 ratio of a Wide camera having a diagonal FOV.sub.W=82°. DA is the aperture diameter. For cut lenses, an effective aperture diameter is given. "Effective aperture diameter" means here a diameter of a circular (or axial symmetric) aperture, wherein the circular aperture has a same aperture area as the cut lens (which has a non axial-symmetric aperture). EFL.sub.G1 and EFL.sub.G2 are the effective focal lengths of lens groups G1 and G2 respectively. The average lens thickness ("ALT") measures the average thickness of all lens elements. ALT.sub.G1 and ALT.sub.G2 is the ALT of G1 and G2 respectively. T.sub.1 is the center thicknesses of L.sub.1. F.sub.1 is the focal length of L.sub.1. All other parameters not specifically defined here have their ordinary meaning as known in the art.

(119) TABLE-US-00014 TABLE 14 Parameter 800 900 1000 1100 Explanation Type P P P A-P
 Sensor configuration H.sub.G1 6.00 5.00 5.00 5.10 Optical height of G1 W.sub.G1 8.00 6.00 6.00 6.00 Optical width of G1 H.sub.G2 4.40 — — — Optical height of G2 W.sub.G2 5.56 — — — Optical width of G2 Δ C 0.15 0.15 0.15 0.00 Lens-OPFE de-center HA 6.00 5.00 5.00 5.10 Aperture height of lens WA 8.00 6.00 6.00 6.00 Aperture width of lens DA 7.40 5.75 5.75 5.80 Aperture diameter of lens HA.sub.G2 4.40 — — — HA of G2 WA.sub.G2 5.56 — — — WA of G2 DA.sub.G2 5.24 — — — DA of G2 H.sub.P 6.80 6.80 6.80 6.11 Height of prism W.sub.P 13.50 11.00 11.00 10.30 Width of prism L.sub.P 7.70 7.70 7.70 6.72 Length of prism Δ LO 2.70 2.70 2.70 3.10 Distance lens-OPFE TTL 18.72 12.41 14.52 13.68 BFL 5.84 1.50 3.93 2.94 EFL 17.37 14.10 14.10 14.10 EFL.sub.G1 21.47 — — — EFL of G1 EFL.sub.G2 1969.60 — — — EFL of G2 f number 2.35 2.45 2.45 2.43 n-FOV.sub.T 25.6° 22.8° 31° 27.40° Diagonal n-FOV.sub.T s-

FOV.sub.T 50.9° × 69.5° × 69.5° × 69.4° × 32.5° 42.58° 42.58° 42.6° 16:9 W ratio N Y Y Y α-
 OPFE ±7.85° ±11.67 ±9.67 ±13.18 Maximal rotation around 1.sup.st rotation axis (402, 452, 1306)
 β-OPFE ±21.67° ±29.80 ±25.54 ±21.35 Maximal rotation around 2.sup.nd rotation axis (502, 552,
 1312) SD 8.00 5.60 8.00 7.00 Image sensor diagonal H.sub.sensor 4.80 1.68 2.40 5.60 Sensor
 height MH.sub.M 8.24 8.84 8.48 7.48 Minimum module height MH.sub.S 6.16 5.95 6.18 5.34
 Minimum shoulder height H.sub.M 9.74 10.34 9.98 8.98 Module height H.sub.S 7.66 7.45 7.68
 6.84 Shoulder height ALT 1.74 0.82 1.33 1.00 Average thickness of lens elements L1-L6
 ALT.sub.G1 2.72 — — — ALT of G1 ALT.sub.G2 1.25 — — — ALT of G2 T.sub.1 3.00 1.94 2.71
 1.60 Center thickness of L.sub.1 f.sub.1 10.42 7.62 9.05 9.56 Focal length of L.sub.1 ΔC/HA 0.025
 0.030 0.030 0.000 ΔC/H.sub.S 0.02 0.020 0.020 0.000 ΔC/Ho 0.022 0.022 0.022 0.000 D-cut ratio
 0.75 0.83 0.83 0.85 =HA/WA (Lens, G1) D-cut ratio 0.79 — — — =H.sub.G2/W.sub.G2 (G2) D-
 cut ratio 0.88 0.88 0.88 0.91 =H.sub.0/L.sub.0 (OPFE) EFL/TTL 0.93 1.14 0.97 1.03 BFL/EFL
 0.34 0.11 0.28 0.21 BFL/TTL 0.31 0.12 0.27 0.21 DA/DA.sub.G2 1.41 — — — DA/H.sub.S 0.97
 0.77 0.75 0.85 WA/H.sub.S 1.04 0.81 0.78 0.88 DA/H.sub.M 0.76 0.56 0.58 0.65
 H.sub.G1/H.sub.G2 1.37 — — — H.sub.G1/MH.sub.S 0.97 0.84 0.81 0.96 H.sub.G1/MH.sub.M
 0.73 0.57 0.59 0.68 H.sub.G1/H.sub.S 0.78 0.67 0.65 0.75 H.sub.G1/H.sub.M 0.62 0.48 0.50 0.57
 H.sub.S/H.sub.M 0.79 0.72 0.77 0.76 SD/EFL 0.46 0.40 0.57 0.50 T.sub.1/ALT 1.73 2.37 2.04 1.60
 ALT.sub.G1/ALT 1.56 — — — ALT.sub.G2/ALT 0.72 — — — ALT.sub.G1/ALT.sub.G2 2.17 —
 — — f.sub.1/EFL 0.60 0.54 0.64 0.68 ΔLO/TTL 0.14 0.22 0.19 0.23

(120) While this disclosure has been described in terms of certain embodiments and generally associated methods, alterations and permutations of the embodiments and methods will be apparent to those skilled in the art. The disclosure is to be understood as not limited by the specific embodiments described herein, but only by the scope of the appended claims.

(121) Furthermore, for the sake of clarity the term “substantially” is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term “substantially” used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to another example, the term “substantially” used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value. According to a further example, the term “substantially” used herein should be interpreted to imply possible variation of up to 1% over or under any specified value.

(122) All references mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual reference was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

Claims

1. A scanning Tele camera (STC), comprising: a single optical path folding element (OPFE) for folding a first optical path OP1 that defines a height H axis to a second optical path OP2 that defines a length L axis; an OPFE actuator; a lens having a lens optical axis parallel to OP2, an effective focal length EFL in the range of 8 mm-50 mm, and a f number $f/\# < 3.5$; and an image sensor, wherein the STC has a STC native field-of-view n-FOV.sub.T, wherein the OPFE actuator is configured to rotate the OPFE around a first rotation axis and around a second rotation axis for scanning a scene with the n-FOV that provides an effective Tele scanning FOV s-FOV.sub.T, wherein the s-FOV.sub.T has a longer side and a shorter side, and wherein the longer side of s-FOV.sub.T is greater than 40 degrees.
2. The STC of claim 1, wherein the first rotation axis is perpendicular to both OP1 and OP2, and wherein the second rotation axis is parallel to OP1.
3. The STC of claim 1, wherein the OPFE has an OPFE length L.sub.O measured along an axis

- parallel to OP2, an OPFE light entering surface and an OPFE light exiting surface, wherein the first rotation axis is located at a distance $\Delta 1$ from the light exiting surface of the OPFE, and wherein $\Delta 1/L_{sub.O} < 0.25$.
4. The STC of claim 3, wherein $\Delta 1/L_{sub.O} < 0.15$.
 5. The STC of claim 1, wherein $f/\# < 3$.
 6. The STC of claim 1, wherein $f/\# < 2.5$.
 7. The STC of claim 1, wherein the STC is included in a dual-camera that comprises a Wide camera having a Wide camera field of view $FOV_{sub.W}$, and wherein a center location of $s-FOV_{sub.T}$ is identical with a center location of $FOV_{sub.W}$.
 8. The STC of claim 7, wherein $FOV_{sub.W}$ is in the range of 50-120 degrees, and wherein $s-FOV_{sub.T}$ covers a 16:9 segment of $FOV_{sub.W}$.
 9. The STC of claim 7, wherein $FOV_{sub.W}$ is in the range of 75-85 degrees, and wherein $s-FOV_{sub.T}$ covers a 16:9 segment of $FOV_{sub.W}$.
 10. The STC of claim 1, wherein the longer side of $s-FOV_{sub.T}$ is greater than 45 degrees.
 11. The STC of claim 1, wherein the longer side of $s-FOV_{sub.T}$ is greater than 50 degrees.
 12. The STC of claim 1, wherein the shorter side of $s-FOV_{sub.T}$ is greater than 20 degrees.
 13. The STC of claim 1, wherein the shorter side of $s-FOV_{sub.T}$ is greater than 25 degrees.
 14. The STC of claim 1, wherein the shorter side of $s-FOV_{sub.T}$ is greater than 30 degrees.
 15. The STC of claim 1, wherein the rotation of the OPFE along the first rotation axis is by more than ± 5 degrees around a zero scan position.
 16. The STC of claim 1, wherein the rotation of the OPFE along the second rotation axis is by more than ± 15 degrees around a zero scan position.
 17. The STC of claim 1, wherein the OPFE is a prism.
 18. The STC of claim 1, wherein the lens has a maximum lens aperture height $H_{sub.A}$ measured along OP1, wherein the STC is included in a camera module, wherein the camera module is divided into a module region having a module region height $H_{sub.M}$ and a shoulder region having a shoulder region height $H_{sub.S} < H_{sub.M}$, all heights being measured along OP1, and wherein $H_{sub.S} < H_{sub.A} + 3$ mm.
 19. The STC of claim 1, wherein the STC is included in a mobile device.
 20. The STC of claim 19, wherein the mobile device is a smartphone.
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